Maintenance Schedule

Diesel Engine
V4000R43x
Application group 2A

MS50140/05E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
8V4000R43	125 kW/cyl.	2A continuous operation, unrestricted
12V4000R43	125 kW/cyl.	2A continuous operation, unrestricted
16V4000R43	138 kW/cyl.	2A continuous operation, unrestricted
20V4000R43	135 kW/cyl.	2A continuous operation, unrestricted
8V4000R43L	150 kW/cyl.	2A continuous operation, unrestricted
12V4000R43L	150 kW/cyl.	2A continuous operation, unrestricted
16V4000R43L	150 kW/cyl.	2A continuous operation, unrestricted
20V4000R43L	150 kW/cyl.	2A continuous operation, unrestricted
16V4000R43R	125 kW/cyl.	2A continuous operation, unrestricted

Table 1: Applicability

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1.15 Maintenance Schedule Matrix 18000 HRS 46
1.16 Cyclical maintenance tasks 18000 HRS 50

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1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer Rolls-Royce Solutions.

Maintenance concept

The maintenance system for Rolls-Royce Solutions products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of Rolls-Royce Solutions. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
QL3: Maintenance work which requires partial disassembly of the product.
QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the product is to remain out of service for a longer time period, Rolls-Royce Solutions recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (HRS)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

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1.3 Allocation matrix application group 2A

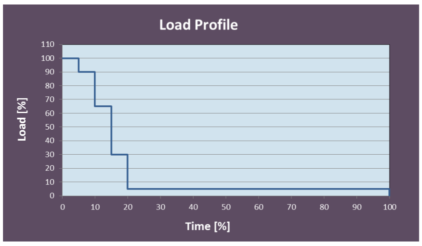
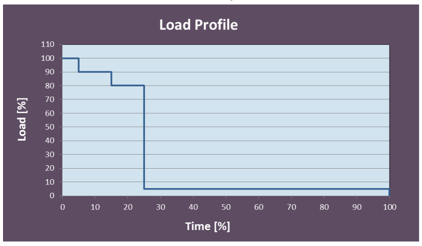
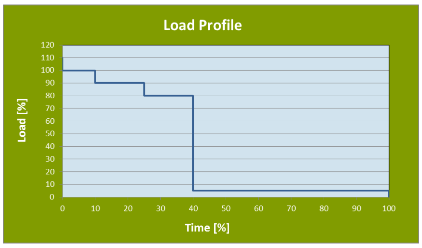
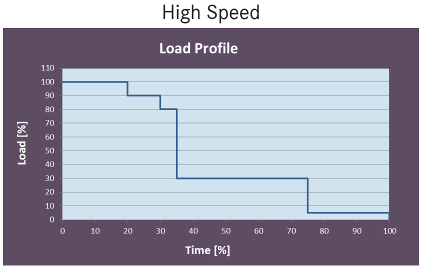
Load profile: Load %	Load profile: Time %	Load factor %	Load indicator %	0 to 125 kW/cyl.	126 to 138 kW/cyl.	139 to 150 kW/cyl.		
110 (100-110)	–	0 to 18	≤ 7	36,000	36,000	30,000	–	Shunter 
100 (90-100)	5							
90 (80-90)	5							
80 (65-80)	–							
65 (50-65)	5							
50 (30-50)	–							
30 (5-30)	5							
Idle (0-5)	80							
110 (100-110)	–	19 to 26	≤ 10	36,000	30,000	27,000	–	Multi Purpose 
100 (90-100)	5							
90 (80-90)	10							
80 (65-80)	10							
65 (50-65)	–							
50 (30-50)	–							
30 (5-30)	–							
Idle (0-5)	75							
110 (100-110)	–	27 to 38	≤ 17	30,000	27,000	24,000	–	Standard Loco 
100 (90-100)	10							
90 (80-90)	15							
80 (65-80)	15							
65 (50-65)	–							
50 (30-50)	–							
30 (5-30)	–							
Idle (0-5)	60							
110 (100-110)	–	39 to 46	≤ 25	27,000	24,000	21,000	–	High Speed 
100 (90-100)	20							
90 (80-90)	10							
80 (65-80)	5							
65 (50-65)	–							
50 (30-50)	–							
30 (5-30)	40							
Idle (0-5)	25							

Table 2: Allocation matrix application group 2A

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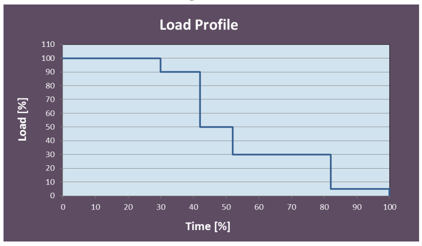
Load profile		Load factor %	Load indicator %	0 to 125 kW/cyl.	126 to 138 kW/cyl.	139 - 150 kW/cyl.		
Load %	Time %							
110 (100-110)	–	47 to 56	≤ 35	24,000	21,000	18,000	–	High Load 
100 (90-100)	30							
90 (80-90)	12							
80 (65-80)	–							
65 (50-65)	–							
50 (30-50)	10							
30 (5-30)	30							
Idle (0-5)	18							
Max. number of load cycles 1				(<10% to >90% set value): 5/h				
Max. number of load cycles 2				(<10% to >70..90% set value): 10/h				
Max. number of starts				2 per engine operating hour				

Table 3: Allocation matrix application group 2A

TIM-ID: 0000044604 - 003

1.4 Noncyclical maintenance tasks

No.	Qual-ification	Task	Maintenance tasks	Reference task	Off-set	Value	Unit	Option
0001	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul.		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

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1.5 Maintenance Schedule Matrix 36000 HRS

500 to 15000 Operating hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00285	1 DAY																														
WM00286	1 DAY																														
WM00287	1 DAY																														
WM00288	1 DAY																														
WM00289	1 DAY																														
WM00290	1 DAY																														
WM00321	1 MON																														
WM00881	1 MON																														
WM00126	1 YR																														
WM00128	1 YR																														
WM00190	1 YR																														
WM00191	1 YR																														
WM00067	2 YR																														
WM00063	2 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00066	9 YR	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00149	1 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00069	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00084	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00085	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00086	2 YR		X		X		X		X		X		X		X		X		X		X		X		X		X		X		X
WM00050	1 YR						X						X						X						X						X
WM00136	1 YR						X						X						X						X						X
WM00188	1 YR						X						X						X						X						X
WM00189	1 YR						X						X						X						X						X
WM00076	3 YR						X						X						X						X						X
WM00664	3 YR						X						X						X						X						X
WM00052	18 YR						X						X						X						X						X
WM00094	18 YR		X				X						X						X						X						X
WM00106	3 YR																		X												
WM00089	18 YR																		X												
WM00090	18 YR																		X												
WM00098	18 YR																		X												
WM00110	18 YR																		X												
WM00775	18 YR																								X						
WM01555	18 YR																														
WM00091	18 YR																								X						
WM00976	18 YR																								X						
WM00008	6 YR																														
WM00065	9 YR																														
WM00054	18 YR																														
WM00056	18 YR																														

DAY = Days
MON = Months
YR = Years
HRS = Hours

Task	Limit	Operating hours [HRS]																													
		500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500	7000	7500	8000	8500	9000	9500	10000	10500	11000	11500	12000	12500	13000	13500	14000	14500	15000
WM00062	18 YR																														
WM00093	18 YR																														
WM00096	18 YR																														
WM00164	18 YR																														
WM00921	18 YR																														
WM00970	18 YR																														
WM00971	18 YR																														
WM00973	18 YR																														
WM00974	18 YR																														
WM00981	18 YR																														
WM00986	18 YR																														
WM00009	15 YR																														
WM00006	6 YR																														
WM00001	18 YR																														
WM00002	18 YR																														
WM00011	18 YR																														
WM00020	18 YR																														
WM00045	18 YR																														
WM00061	18 YR																														
WM00070	18 YR																														
WM00072	18 YR																														
WM00087	18 YR																														
WM00088	18 YR																														
WM00092	18 YR																														
WM00095	18 YR																														
WM00099	18 YR																														
WM00103	18 YR																														
WM00104	18 YR																														
WM00108	18 YR																														
WM00111	18 YR																														
WM00127	18 YR																														
WM00137	18 YR																														
WM00139	18 YR																														
WM00168	18 YR																														
WM00353	18 YR																														
WM00439	18 YR																														
WM00925	18 YR																														
WM00980	18 YR																														

DAY = Days
MON = Months
YR = Years
HRS = Hours

TIM-ID: 0000000000 - 000